

Investment Operations Platform

**Trade Processing, Reconciliation, Position Management &
Portfolio Reporting**

Prepared By

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1. Executive Summary

This project was developed to understand the operational workflow involved in investment management and portfolio operations. The platform simulates the lifecycle of a trade from execution through reconciliation, position management, portfolio valuation, profit and loss reporting and dashboard visualization. Python was used for automation and calculations, Excel for reporting outputs and Power BI for dashboard development.

Project Deliverables

- Trade Generation Engine
- Broker Confirmation File
- Reconciliation Engine
- Exception Reporting
- Position Engine
- Portfolio Valuation Engine
- Profit & Loss Engine
- Power BI Dashboard

2. Workflow Architecture

Trades.xlsx



Broker.xlsx



Reconciliation Engine



Exception Report



Position Engine



Closing Positions



Pricing Engine



Portfolio Valuation



P&L; Engine



KPI Reports



Power BI Dashboard

3. Reconciliation Engine

The reconciliation engine compares internally generated trades against broker records and identifies operational exceptions. Three types of checks were implemented:

- Missing Trade Detection
- Quantity Mismatch Detection
- Price Mismatch Detection

Reconciliation Results

Total Trades Processed: 500

Exceptions Identified: 24

Quantity Mismatch: 10

Price Mismatch: 9

Missing in Broker: 5

4. Portfolio Valuation & P&L;

Portfolio positions were valued using generated market prices. The valuation process included:

- Average Cost Calculation
- Position Cost Calculation
- Market Value Calculation
- Unrealized Profit & Loss Calculation
- Portfolio KPI Reporting

Portfolio Metrics

Total Portfolio Value: 31.76 Million

Total Position Cost: 31.10 Million

Total Unrealized P&L;: 668.84 Thousand

5. Key Learnings

- Understanding trade lifecycle workflows
- Broker reconciliation processes
- Exception management and controls
- Position management calculations
- Portfolio valuation techniques
- Profit & Loss reporting
- Power BI dashboard development
- Python automation using Pandas and NumPy

Project Conclusion

The project successfully simulated key investment operations workflows including trade processing, reconciliation, position management, portfolio valuation and performance reporting. The final outputs were integrated into a Power BI dashboard to provide portfolio and operational insights.